Lecture 11. Causal Inference

Potential outcomes, randomization, confounding, matching and weighting

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11.1 Why causation

Motivation. Causal questions ask what Y would be if we changed something; answering needs assumptions on treatment assignment.

Causal questions: is smoking harmful? Would cleaner air reduce respiratory deaths? A **confounder** X affects both treatment Z and outcome Y (hot weather drives ice-cream sales and drownings).

Job training (LaLonde). Did a 1976 programme raise 1978 earnings? The groups differ before treatment:

```
lalonde %>%
  group_by(treat) %>%
  summarise(across(c(age, educ, married, re74, re75, re78), mean), black = mean(race ==
    "black"))
```

```
## # A tibble: 2 x 8
##   treat   age  educ married  re74  re75  re78 black
##   <int> <dbl> <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
## 1     0  28.0  10.2   0.513 5619. 2466. 6984. 0.203
## 2     1  25.8  10.3   0.189 2096. 1532. 6349. 0.843
```

The treated are more often Black, less often married, and earned far less in 1974–75.

Smoking and lung cancer. Lung cancer tripled by the mid-1940s; observational studies followed in 1950 (Wynder and Graham; Doll and Hill; Hammond). Hammond matched 36 975 heavy smokers to non-smokers on many covariates. In 122 discordant pairs the smoker died in 110; with no effect $X \sim B(122, 1/2)$:

```
binom.test(110, 122, p = 0.5, alternative = "greater")$p.value
```

```
## [1] 2.733658e-21
```

The weak point is an *unmeasured* confounder (Fisher: a gene causing both).

Counterfactuals. Zeus dies five days after a heart transplant; had he lived without it, the transplant caused his death. Causal effects compare what happened with what *would have happened*.

11.2 Potential outcomes

Motivation. Potential outcomes give notation in which both treated and untreated outcomes exist.

For units $i = 1, \ldots, N$: covariates X_i; treatment $Z_i \in \{0, 1\}$; **potential outcomes** $Y_i(1)$ (if treated) and $Y_i(0)$ (if not); and the observed outcome

$$Y_i = Y_i(Z_i) = Z_i Y_i(1) + (1 - Z_i) Y_i(0).$$

For Zeus, $Z_i = 1$ and $Y_i = 1$ (death), with $Y_i(1) = 1$, $Y_i(0) = 0$: $Y_i(1)$ is *factual*, $Y_i(0)$ *counterfactual*. (Also written treatment A and potential outcomes $Y^{a=1}$, $Y^{a=0}$.)

Hidden assumptions:

- **no interference**: Y_i depends only on Z_i;
- **consistency**: $Y_i = Y_i(1)$ if $Z_i = 1$ (**no multiple versions of treatment**).

Together these are the **stable unit treatment value assumption** (SUTVA).

Causal effects. The individual effect is $\tau_i = Y_i(1) - Y_i(0)$; the **average treatment effect** is

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)].$$

Other measures: the risk ratio $\mathbb{E}[Y_i(1)]/\mathbb{E}[Y_i(0)]$ and, for binary outcomes, the odds ratio $\frac{P(Y_i(1)=1)\,P(Y_i(0)=0}{P(Y_i(1)=0)\,P(Y_i(0)=1}$

The **sample** ATE is $\frac{1}{N}\sum_i\{Y_i(1)-Y_i(0)\}$, the **population** ATE is $\mathbb{E}[Y_i(1)-Y_i(0)]$, and SATE $\to$ PATE as $N\to\infty$.

Science table. Both potential outcomes per unit; the data show one, the other is "?".

The fundamental problem of causal inference (Holland, 1986): τ_i is never observed. The ATE can be identified under random assignment or unconfoundedness, with **overlap** $0 < P(Z = 1 \mid X) < 1$.

11.3 Randomised experiments and observational studies

Motivation. Randomisation balances everything; without it, comparable groups must be built from covariates.

Randomised experiments (trials, A/B tests) are the gold standard but often infeasible; observational data are easy to get, hard to analyse. Randomisation balances observed and unobserved covariates and is the "reasoned basis for inference" (Fisher, 1935):

$$(Y(1), Y(0)) \perp\!\!\!\perp Z.$$

Then $\mathbb{E}[Y(1)] = \mathbb{E}[Y(1) \mid Z = 1] = \mathbb{E}[Y(1) \mid Z = 0]$, and $Y(1)$ is observed when $Z = 1$, so

$$\mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1) \mid Z = 1] - \mathbb{E}[Y(0) \mid Z = 0] = \mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0] = \bar{Y}_t - \bar{Y}_c,$$

with no distributional assumption (the **identification assumption** is the independence).

When it fails. If a doctor treats patients with high expected $Y_i(0)$, then $\mathbb{E}[Y(0) \mid Z = 1] > \mathbb{E}[Y(0) \mid Z = 0]$: the medical history X confounds. **Overt bias** (observed X) is removed by regression, matching, weighting; **hidden bias** needs special designs (instruments, discontinuities, difference-in-differences) and sensitivity analysis.

11.3.1 Unconfoundedness

The key assumption is

$$(Y(1), Y(0)) \perp\!\!\!\perp Z \mid X \qquad \text{(unconfoundedness, ignorability)}:$$

treatment is random among units with the same X. Then

$$\mathbb{E}[Y(1)] = \mathbb{E}\big[\mathbb{E}[Y(1) \mid X]\big] = \mathbb{E}\big[\mathbb{E}[Y(1) \mid X, Z = 1]\big] = \mathbb{E}\big[\mathbb{E}[Y \mid X, Z = 1]\big],$$

and likewise for $Y(0)$. With outcome models $\hat{\mu}_1$ (treated) and $\hat{\mu}_0$ (controls),

$$\hat{\tau}^{\text{ATE}} = \frac{1}{n}\sum_{i=1}^{n}\{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}.$$

In general $\mathbb{E}[Y(1)] \neq \mathbb{E}[Y \mid Z = 1]$.

Connection to regression. The classical adjustment $\mathbb{E}(Y \mid X) = \beta_0 + \beta_1 Z + \beta_2 X$ corresponds to $\mathbb{E}[Y(1) \mid X] = \beta_0 + \beta_1 + \beta_2 X$ and $\mathbb{E}[Y(0) \mid X] = \beta_0 + \beta_2 X$, so the ATE is β_1. But the two means may need different forms, $\tau(X)$ may vary with X, poor overlap means *extrapolation*, and adjusting while seeing outcomes invites cherry-picking.

11.4 Propensity score methods

Motivation. The propensity score compresses many covariates into one number that suffices for comparison.

The **propensity score** $e(X_i) = P(Z_i = 1 \mid X_i)$ (Rosenbaum and Rubin, 1983):

Theorem 11.1 (balancing property)

If $(Y(1), Y(0)) \perp\!\!\!\perp Z \mid X$, then $(Y(1), Y(0)) \perp\!\!\!\perp Z \mid e(X)$.

Proof [Supplement]. It suffices to show $P(Z = 1 \mid Y(1), Y(0), e(X)) = e(X)$, which does not depend on the potential outcomes. By the tower property, $P(Z = 1 \mid Y(1), Y(0), e(X)) = \mathbb{E}[P(Z = 1 \mid Y(1), Y(0), X) \mid Y(1), Y(0), e(X)] = \mathbb{E}[e(X) \mid Y(1), Y(0), e(X)] = e(X)$, using unconfoundedness in the middle step and that $e(X)$ is a function of X. ■

Sharing $e(X)$ is enough. In observational studies it is estimated, usually by logistic regression, $\operatorname{logit} P(Z_i = 1 \mid X_i) = \beta_X^\mathsf{T} X_i$ and $\hat{e}(X_i) = \operatorname{expit}(\hat{\beta}_X^\mathsf{T} X_i)$. then used to stratify, match or weight.

11.4.1 Stratification

(1) Estimate $\hat{e}(X_i)$; (2) cut it into K narrow strata (quantiles or $(0, \frac{1}{K}], \ldots, (\frac{K-1}{K}, 1)$); (3) treat each stratum as a randomised experiment:

$$\hat{\tau} = \sum_{k=1}^{K} (\bar{Y}_{k1} - \bar{Y}_{k0}) \, \frac{n_k}{n},$$

with $\bar{Y}_{k1}$, $\bar{Y}_{k0}$ the treated and control means in stratum k of size n_k. Equivalently, a treated unit in stratum k has weight $\frac{1}{n_{k1}} \frac{n_k}{n}$ and a control $\frac{1}{n_{k0}} \frac{n_k}{n}$, and a weighted regression $Y = \alpha + \beta Z + \epsilon$ gives $\hat{\tau} = \hat{\beta}$.

11.4.2 Matching

1:1 matching (fewer treated than controls): (1) match each treated unit to a control with similar X or $e(X)$, with or without replacement (bias–variance); (2) check balance; (3) discard unmatched controls; (4) analyse as a paired experiment: mean treated-minus-control difference (or regress Y on Z), randomisation test of no effect.

11.4.3 Inverse probability weighting

IPW weights treated units by $1/e(X)$, controls by $1/(1-e(X))$: a treated person with $e(X) = 1/5$ stands for five "clones". The **pseudo-population** has $(Y(1), Y(0)) \perp\!\!\!\perp Z$.

Proposition 11.2 (IPW identification)

Under unconfoundedness and overlap,

$$\tau = \mathbb{E}\Big[\frac{Z_i Y_i}{e(X_i)}\Big] - \mathbb{E}\Big[\frac{(1 - Z_i) Y_i}{1 - e(X_i)}\Big].$$

Proof. First term: $\mathbb{E}[\frac{Z_i Y_i}{e(X_i)}] = \mathbb{E}[\frac{\mathbb{E}[Z_i Y_i \mid X_i]}{e(X_i)}] = \mathbb{E}[\frac{e(X_i)\,\mathbb{E}[Y_i \mid Z_i = 1, X_i]}{e(X_i)}] = \mathbb{E}[\mathbb{E}[Y_i(1) \mid Z_i = 1, X_i]] = \mathbb{E}[\mathbb{E}[Y_i(1) \mid X_i]] = \mathbb{E}[Y_i(1)]$, using $\mathbb{E}[Z_i Y_i \mid X_i] = P(Z_i = 1 \mid X_i)\mathbb{E}[Y_i \mid Z_i = 1, X_i]$, consistency and unconfoundedness. The second term equals $\mathbb{E}[Y_i(0)]$ in the same way [Supplement]. ■

With estimated scores $\hat{e}_i$,

$$\hat{\tau}^{\text{ATE}} = \frac{1}{n}\sum_i \frac{Z_i Y_i}{\hat{e}_i} - \frac{1}{n}\sum_i \frac{(1-Z_i)Y_i}{1-\hat{e}_i},$$

or a regression of Y on Z with weights $w_i = Z_i/\hat{e}_i + (1-Z_i)/(1-\hat{e}_i)$, unstable for $\hat{e}_i$ near 0 or 1. ATT weights: 1 (treated), $\hat{e}_i/(1-\hat{e}_i)$ (controls; Exercise 11.3).

11.5 Example: job training and earnings

Packages: `MatchIt`, `WeightIt`; balance by `cobalt::bal.tab()`, effects by `marginaleffects::avg_comparisons()`. Data: `cobalt::lalonde` (185 treated, 429 controls).

```
no.adj <- matchit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                  data = lalonde, method = NULL, distance = "glm")
summary(no.adj)$sum.all[, 1:3]
```

```
##            Means Treated Means Control Std. Mean Diff.
## distance    5.774355e-01     0.1822248      1.79408605
## age         2.581622e+01    28.0303030     -0.30944526
## educ        1.034595e+01    10.2354312      0.05496466
## raceblack   8.432432e-01     0.2027972      1.76154189
## racehispan  5.945946e-02     0.1421911     -0.34984254
## racewhite   9.729730e-02     0.6550117     -1.88186760
## married     1.891892e-01     0.5128205     -0.82630927
## nodegree    7.081081e-01     0.5967366      0.24497023
## re74        2.095574e+03  5619.2365064     -0.72108381
## re75        1.532055e+03  2466.4844431     -0.29026291
```

The standardised mean differences are huge (1.76 Black, −1.88 White, −0.72 1974 earnings); a rule of thumb is $|\text{SMD}| < 0.1$ [Supplement]. Three adjustments:

```
ps.str <- matchit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                  data = lalonde, method = "subclass", distance = "glm", link = "probit"
    , subclass = 7)
ps.mat <- matchit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                  data = lalonde, method = "nearest", distance = "glm", link = "probit",
     replace = TRUE)
W <- weightit(treat ~ age + educ + race + married + nodegree + re74 + re75,
              data = lalonde, method = "glm", estimand = "ATT")
round(cbind(stratified = summary(ps.str, un = FALSE)$sum.across[, 3],
            matched = summary(ps.mat, un = FALSE)$sum.matched[, 3]), 3)
```

```
##            stratified matched
## distance        0.063   0.003
## age            -0.036  -0.085
## educ            0.027  -0.083
## raceblack       0.084   0.015
## racehispan     -0.018   0.046
## racewhite      -0.089  -0.055
## married        -0.030   0.097
## nodegree        0.027   0.143
## re74           -0.089  -0.013
## re75           -0.074  -0.017
```

```
bal.tab(W, un = TRUE)
```

```
## Balance Measures
##                 Type Diff.Un Diff.Adj
## prop.score  Distance  1.7941  -0.0205
## age          Contin. -0.3094   0.1188
## educ         Contin.  0.0550  -0.0284
## race_black    Binary  0.6404  -0.0022
## race_hispan   Binary -0.0827   0.0002
```

```
## race_white    Binary -0.5577   0.0021
## married       Binary -0.3236   0.0186
## nodegree      Binary  0.1114   0.0184
## re74         Contin. -0.7211  -0.0021
## re75         Contin. -0.2903   0.0110
##
## Effective sample sizes
##            Control Treated
## Unadjusted  429.       185
## Adjusted     99.82     185
```

All three nearly balance (7 strata; matching keeps 84 distinct controls; IPW keeps everyone). ATT by weighted regression of 1978 earnings on treatment:

```
str.data <- match.data(ps.str); mat.data <- match.data(ps.mat)
est <- function(fit, ...) {
  a <- avg_comparisons(fit, variables = "treat", ...)
  round(c(estimate = a$estimate, low = a$conf.low, high = a$conf.high))
}
fit.str <- lm(re78 ~ treat, data = str.data, weights = weights)
fit.mat <- lm(re78 ~ treat, data = mat.data, weights = weights)
lalonde$ipw.weight <- W$weights
fit.ipw <- lm(re78 ~ treat, data = lalonde, weights = ipw.weight)
rbind(
  stratification = est(fit.str, vcov = ~subclass, newdata = subset(str.data, treat == 1)
    , wts = "weights"),
  matching       = est(fit.mat, vcov = "HC3", newdata = subset(mat.data, treat == 1),
    wts = "weights"),
  ipw            = est(fit.ipw, vcov = "HC3", wts = "ipw.weight"),
  naive          = round(c(coef(lm(re78 ~ treat, data = lalonde))[2], NA, NA)))
```

```
##                estimate  low high
## stratification     1301 -423 3025
## matching           1290 -603 3184
## ipw                1214 -416 2844
## naive              -635   NA   NA
```

Full matching (no units discarded):

```
ps.full <- matchit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                   data = lalonde, method = "full", distance = "glm", link = "probit")
full.data <- match.data(ps.full)
fit.full <- lm(re78 ~ treat, data = full.data, weights = weights)
est(fit.full, vcov = ~subclass, newdata = subset(full.data, treat == 1), wts = "weights"
    )
```

```
## estimate      low     high
##     1855      364     3346
```

Naive: the programme *lowered* earnings; adjusted: about \$1 200–1 900. The original experiment found \$2 114, 95% CI (458, 3955), closest to full matching.

Take-home. Matching and weighting separate *design* (no outcomes) from *analysis* (compare means); try several methods and check agreement.

11.6 Selected exercises

Motivation. Why the naive comparison had the wrong sign and why the adjustments agree.

Throughout, consistency and SUTVA hold, $e(X) = P(Z = 1 \mid X)$, and expectations are finite.

Exercise 11.1 (the naive comparison)

Show that

$$\mathbb{E}[Y \mid Z=1] - \mathbb{E}[Y \mid Z=0] = \underbrace{\mathbb{E}[Y(1)-Y(0) \mid Z=1]}_{\text{ATT}} + \underbrace{\mathbb{E}[Y(0) \mid Z=1] - \mathbb{E}[Y(0) \mid Z=0]}_{\text{selection bias}},$$

that the bias vanishes and ATT = ATE under random assignment, and interpret the sign of the bias in the job-training data.

Solution [Supplement].

Step 1. By consistency, $\mathbb{E}[Y \mid Z=1] = \mathbb{E}[Y(1) \mid Z=1]$ and $\mathbb{E}[Y \mid Z=0] = \mathbb{E}[Y(0) \mid Z=0]$. Adding and subtracting $\mathbb{E}[Y(0) \mid Z=1]$ gives the decomposition.

Step 2. If $(Y(1), Y(0)) \perp\!\!\!\perp Z$, then $\mathbb{E}[Y(0) \mid Z=1] = \mathbb{E}[Y(0) \mid Z=0] = \mathbb{E}Y(0)$, so the bias is 0, and likewise $\mathbb{E}[Y(1)-Y(0) \mid Z=1] = \mathbb{E}[Y(1)-Y(0)]$.

Step 3. Naive difference negative, adjusted ATT +\$1 200 to +\$1 900: the selection bias is strongly negative (participants would have earned much less anyway, as 1974–75 earnings show). ■

Exercise 11.2 (stratification is weighting)

Divide n units into strata $k = 1, \dots, K$ of sizes n_k, with $n_{k1} \geq 1$ treated and $n_{k0} \geq 1$ control units, and estimate the propensity of every unit in stratum k by $\hat{e}_k = n_{k1}/n_k$. Show that the IPW estimator

$$\hat{\tau}^{\text{IPW}} = \frac{1}{n}\sum_i \frac{Z_i Y_i}{\hat{e}_i} - \frac{1}{n}\sum_i \frac{(1-Z_i)Y_i}{1-\hat{e}_i}$$

equals the stratified estimator $\sum_k (\bar{Y}_{k1} - \bar{Y}_{k0})\, n_k/n$ of Section 11.4.

Solution [Supplement]. Group the sums by stratum. For the treated part,

$$\frac{1}{n}\sum_k \frac{n_k}{n_{k1}} \sum_{i \in k,\, Z_i=1} Y_i = \frac{1}{n}\sum_k n_k \bar{Y}_{k1} = \sum_k \frac{n_k}{n}\bar{Y}_{k1}.$$

For the control part, $1-\hat{e}_k = n_{k0}/n_k$, and the same computation gives $\sum_k (n_k/n)\bar{Y}_{k0}$. Subtracting yields the stratified estimator: IPW with propensity estimated by the stratum's treated share. ■

Exercise 11.3 (weights for the effect on the treated)

Assume $Y(0) \perp\!\!\!\perp Z \mid X$, $e(X) < 1$ and $P(Z=1) > 0$. Prove that

$$\mathbb{E}[Y(0) \mid Z=1] = \frac{\mathbb{E}\big[(1-Z)\,\frac{e(X)}{1-e(X)}\,Y\big]}{P(Z=1)},$$

and deduce that the ATT is the difference between the mean outcome of the treated (weight 1) and the weighted mean outcome of the controls with weights $e(X)/(1-e(X))$.

Solution [Supplement].

Step 1. Condition on X. By consistency $(1-Z)Y = (1-Z)Y(0)$, and by unconfoundedness

$$\mathbb{E}\Big[(1-Z)\frac{e(X)}{1-e(X)}Y \,\Big|\, X\Big] = \frac{e(X)}{1-e(X)}\,\mathbb{E}[1-Z \mid X]\,\mathbb{E}[Y(0) \mid X] = e(X)\,\mathbb{E}[Y(0) \mid X].$$

Step 2. Again by unconfoundedness, $e(X)\mathbb{E}[Y(0) \mid X] = \mathbb{E}[Z \mid X]\,\mathbb{E}[Y(0) \mid X] = \mathbb{E}[ZY(0) \mid X]$. Taking expectations, the numerator equals $\mathbb{E}[ZY(0)] = P(Z=1)\,\mathbb{E}[Y(0) \mid Z=1]$, which is the claim.

Step 3. With $w = 1$ for treated units and $w = e(X)/(1 - e(X))$ for controls, the same argument with Y replaced by 1 gives $\mathbb{E}[(1-Z)w] = P(Z = 1)$, so $\mathbb{E}[Y(0) \mid Z = 1] = \mathbb{E}[(1-Z)wY]/\mathbb{E}[(1-Z)w]$, a weighted control mean. Since $\mathbb{E}[Y(1) \mid Z = 1] = \mathbb{E}[Y \mid Z = 1]$, the ATT is the treated mean minus this weighted control mean. ■